# Type MS · Embedded systems: small computers inside devices

## Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a)(i) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• Heart rate (1) to measure blood flow/pulse rate (1)<br>• Barometric pressure (1) to determine altitude (1)<br>• Microphone (1) for voice activation control (1)<br>• Temperature/Thermometer (1) to measure ambient/body temperature (1)<br>• GPS (1) to calculate longitude/latitude (1)<br>• Light (1) to adjust screen brightness/charge watch battery (1)<br>• Magnetometer (1) to act as a compass (1)<br>• ECG (1) measure heart rhythm (1)<br>• Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1) | Allow any reasonable sensor that might be embedded within a smart watch.<br>Purpose must be linked to the sensor<br>Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark)<br>Allow purpose mentioned in second column if nothing in first column | **2** |
| Question | Answer | Additional Guidance | Mark |
| 5(a)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• It has a low power requirement (1)<br>• It can use a low processor speed (1)<br>• It has built in/integrated storage (1)<br>• It relies on flash memory (1)<br>• It can be mass produced at a low cost per device (1)<br>• It is small (1)<br>• It can work in a large range of operating temperature ranges (1)<br>• It is robust (1)<br>• It has its operating software set by the manufacturer (1)<br>• It is designed to do one particular task (1) | | **2** |

# Type MS · Embedded systems: small computers inside devices

## Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part a.ii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | **The only correct answer is B**<br><br>*A is not correct because low-level languages are not converted into high-level languages*<br>*C is not correct because assemblers translate assembly code to produce machine code but do not execute it*<br>*D is not correct because compilers and interpreters translate high-level languages* | | 1 |
| Question | Answer | Additional Guidance | Mark |
| 5(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1)<br>• It reduces manufacturing costs (1) because of a lower processor specification/expense (1)<br>• Optimise their code for hardware (1) directly access/control registers (1) | Do not accept references to translation time | 2 |
| Question | Answer | Additional Guidance | Mark |
| 5(b)(iii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• Abide by the law / avoid plagiarism / copyright infringements (1)<br>• Adhere to a professional code of conduct (1)<br>• Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1)<br>• Test code thoroughly/do not release code with known errors/vulnerabilities (1)<br>• Respect privacy and security of data (1)<br>• Preserve the environment (1)<br>• Make software accessible/unbiased (1) | | 2 |

# Type MS · Embedded systems: small computers inside devices

## Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part a.ii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(iv) | **The only correct answer is A**<br><br>*B is not correct because 500 KiB is 512,000 bytes*<br>*C is not correct because 0.5 MiB is 524,288 bytes*<br>*D is not correct because 5MB is 5,000,000 bytes* | | **1** |
| **Question** | **Answer** | **Additional Guidance** | **Mark** |
| 5(c) | Indicative content:<br><br>• Can use sensors to take/track readings such as heart rate<br>• Can track location via GPS and record routes taken/distance travelled<br>• Can calculate exercise/movement undertaken such as steps/calories burned<br>• Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data<br>• Can trigger alerts when users are inactive<br>• Can monitor blood oxygen/blood sugar/sleep patterns<br>• Can applaud/give kudos when an achievement/event takes place<br>• Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop<br>• Workouts/alerts can be synced to the watch<br>• Linked apps can perform more in-depth analysis<br>• No guarantee that users will buy into benefits and lead a healthier lifestyle<br>• Monitoring vulnerable people allowing independent living | Allow content referring to other wearable technology which is not a smartwatch | **6** |
| | | **Total for question 5** | **16** |

## Type MS · Embedded systems: small computers inside devices

### Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part a.ii in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1-2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

## Type MS · Embedded systems: small computers inside devices

### Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part a.ii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(ii) | Any **two** from:<br>• Printer (1)<br>• Bank card (magnetic strip) reader/scanner (1)<br>• Bank card contactless/NFC reader/scanner (1)<br>• Cash input and giving change device (1) | Allow any hardware that relates to paying by cash for MP4 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(iii) | To prevent unauthorised people from reading / understanding the bank card information (1) | Do not award 'to keep it from being hacked / accessed'. | 1 |

## Type MS · Embedded systems: small computers inside devices

### Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part a.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b) | Award **one** mark for each of (maximum of 4):<br><br>1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1)<br>2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1)<br>3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1)<br>4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1)<br>5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1) | For each mark point:<br>• the general feature is followed by exemplification<br>• needs to be a comparison, which may be implicit | 4 |

## Type MS · Embedded systems: small computers inside devices

### Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part a.i in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|
| 6(c) | Indicative content:<br><br>Characteristics – Any of the following:<br>• Implemented in software.<br>• Could involve the use of neural networks / clustering / modelling<br>• Algorithms are trained by exposure to data that has been validated / checked<br>• Algorithms learn by identifying commonalities between the validated / checked data<br>• Algorithms make predictions for new data based on prior learning<br>• Algorithms may incorporate recursive self-improvement<br><br>Uses – Any of the following:<br>• Game playing (opponents)<br>• Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions)<br>• Image / Audio processing (recognising objects/patterns, face recognition)<br>• Logistics (scheduling, order fulfilment)<br>• Control systems (cars, manufacturing, weapons, navigation<br>• Diagnostic systems (medical, mechanical, electrical)<br>• Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants)<br>• Robotics (dangerous situations, help aged or disabled)<br><br>Ethical issues – Any of the following:<br>• Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes<br>• Is a computer to be trusted to make decisions (life-death)?<br>• If a computer discovers something that humans can't prove, should it be accepted as truth?<br>• How do we know that the answer produced is correct?<br>• Will people be comfortable interacting with machines that are considered intelligent?<br>• Some jobs may be at risk, especially those requiring human skills such as empathy or creativity. | 6 |

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part a.i in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured. |

## Type MS · Embedded systems: small computers inside devices

### Answer 2 of 7 · Question 6 part a.i · 2 marks

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Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(ii) | Any **two** from:<br>• Printer (1)<br>• Bank card (magnetic strip) reader/scanner (1)<br>• Bank card contactless/NFC reader/scanner (1)<br>• Cash input and giving change device (1) | Allow any hardware that relates to paying by cash for MP4 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(a)(iii) | To prevent unauthorised people from reading / understanding the bank card information (1) | Do not award 'to keep it from being hacked / accessed'. | 1 |

## Type MS · Embedded systems: small computers inside devices

### Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part a.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b) | Award **one** mark for each of (maximum of 4):<br><br>1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1)<br>2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1)<br>3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1)<br>4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1)<br>5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1) | For each mark point:<br>• the general feature is followed by exemplification<br>• needs to be a comparison, which may be implicit | 4 |

## Type MS · Embedded systems: small computers inside devices

### Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part a.ii in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|
| 6(c) | Indicative content:<br><br>Characteristics – Any of the following:<br>• Implemented in software.<br>• Could involve the use of neural networks / clustering / modelling<br>• Algorithms are trained by exposure to data that has been validated / checked<br>• Algorithms learn by identifying commonalities between the validated / checked data<br>• Algorithms make predictions for new data based on prior learning<br>• Algorithms may incorporate recursive self-improvement<br><br>Uses – Any of the following:<br>• Game playing (opponents)<br>• Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions)<br>• Image / Audio processing (recognising objects/patterns, face recognition)<br>• Logistics (scheduling, order fulfilment)<br>• Control systems (cars, manufacturing, weapons, navigation<br>• Diagnostic systems (medical, mechanical, electrical)<br>• Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants)<br>• Robotics (dangerous situations, help aged or disabled)<br><br>Ethical issues – Any of the following:<br>• Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes<br>• Is a computer to be trusted to make decisions (life-death)?<br>• If a computer discovers something that humans can't prove, should it be accepted as truth?<br>• How do we know that the answer produced is correct?<br>• Will people be comfortable interacting with machines that are considered intelligent?<br>• Some jobs may be at risk, especially those requiring human skills such as empathy or creativity. | 6 |

## Type MS · Embedded systems: small computers inside devices

### Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part a.ii in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
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## Type MS · Embedded systems: small computers inside devices

### Answer 3 of 7 · Question 6 part a.ii · 2 marks

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## Type MS · Embedded systems: small computers inside devices

### Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a) | Award **one** mark for each correct component:<br><br>**Component** / **Letter**<br>Control unit — B<br>Output device — G<br>Registers — C<br>Main memory — H<br>Cache — F<br>Input device — E<br>Arithmetic/logic unit (ALU) — D<br>Clock — A | | 5 |

## Type MS · Embedded systems: small computers inside devices

### Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | Award **one** mark each to a maximum of **two** marks for:<br><br>• stores a memory address (for the next instruction) (1)<br>• memory address incremented/changed during the fetch process (1) | | 2 |
| 5(b)(ii) | **The only correct answer is B**<br><br>A *is not correct because* there is no such signal<br>C *is not correct because* as there is no such signal<br>D *is not correct because* as that does not take place during the fetch stage | | 1 |
| 5(b)(iii) | Award **one** mark from:<br><br>• processes more instructions in the same amount of time (1)<br>• more cycles can be performed in the same amount of time (1)<br>• fetch-execute-decode cycle runs faster (1) | | 1 |

## Type MS · Embedded systems: small computers inside devices

### Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c)(i) | Award **one** mark from:<br>• a system designed for one/a few specific functions/task(s) (1)<br>• has both hardware and software (1)<br>• has integrated memory (1)<br>• has no or minimal user interface (1)<br>• is power efficient/low power consumption (1)<br>• its functionality cannot be changed/upgraded by users (1)<br>• often use sensors and actuators to interact with the external environment (1)<br>• functions in real time (1) | | 2 |
| 5(c)(ii) | Award up to **two** marks for a linked explanation such as:<br>• Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1)<br>• Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1)<br>• The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1)<br>• Only needs a small bus width (1) because the output signals are serial binary (1) | | 2 |
| | | **Total** | **13** |

## Type MS · Embedded systems: small computers inside devices

### Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Embedded systems: small computers inside devices

### Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a) | Award **one** mark for each correct component:<br><br>**Component** / **Letter**<br>Control unit — B<br>Output device — G<br>Registers — C<br>Main memory — H<br>Cache — F<br>Input device — E<br>Arithmetic/logic unit (ALU) — D<br>Clock — A | | 5 |

## Type MS · Embedded systems: small computers inside devices

### Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | Award **one** mark each to a maximum of **two** marks for:<br><br>• stores a memory address (for the next instruction) (1)<br>• memory address incremented/changed during the fetch process (1) | | 2 |
| 5(b)(ii) | **The only correct answer is B**<br><br>A *is not correct because* there is no such signal<br>C *is not correct because* as there is no such signal<br>D *is not correct because* as that does not take place during the fetch stage | | 1 |
| 5(b)(iii) | Award **one** mark from:<br><br>• processes more instructions in the same amount of time (1)<br>• more cycles can be performed in the same amount of time (1)<br>• fetch-execute-decode cycle runs faster (1) | | 1 |

## Type MS · Embedded systems: small computers inside devices

### Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c)(i) | Award **one** mark from:<br>• a system designed for one/a few specific functions/task(s) (1)<br>• has both hardware and software (1)<br>• has integrated memory (1)<br>• has no or minimal user interface (1)<br>• is power efficient/low power consumption (1)<br>• its functionality cannot be changed/upgraded by users (1)<br>• often use sensors and actuators to interact with the external environment (1)<br>• functions in real time (1) | | 2 |
| 5(c)(ii) | Award up to **two** marks for a linked explanation such as:<br>• Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1)<br>• Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1)<br>• The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1)<br>• Only needs a small bus width (1) because the output signals are serial binary (1) | | 2 |
| | | **Total** | **13** |

## Type MS · Embedded systems: small computers inside devices

### Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

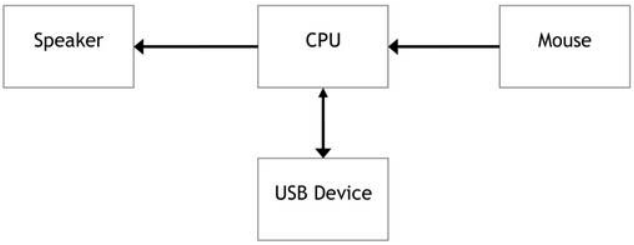
2023 JUN · Paper 01 · extract 1 of 8 · find part c in this mark scheme extract

<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)<br>2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)<br>4. The bee flies to its new position (1)<br>5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.<br><br>The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5b(i) | Award one mark for each:<br>• Line between Mouse and CPU with arrow at CPU end only (1)<br>• Line between CPU and Speaker with arrow at Speaker end only (1)<br>• Line between CPU and USB Device with arrows on both ends (1)<br>  | Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow.<br><br>Ignore any additional lines. | 3 |

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(ii) | **The only correct answer is B**<br><br>***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted*<br><br>***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted*<br><br>***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(iii) | Award up to **two** marks for a linked explanation, such as:<br><br>• Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1)<br><br>• Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1)<br><br>• It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1) | | 2 |

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c) | Award up to **two** marks for a linked description, such as:<br>• Uses a sensor (1) to check the water level in the machine (1)<br>• Uses a thermometer (1) to check the water temperature (1)<br>• Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1)<br>• Uses a motor (1) to rotate the drum (1) | The first mark is for identification and the second is for the linked description.<br><br>It is possible to gain one mark for a description without identifying the sensor/actuator. | 2 |

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.<br><br>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:<br><br>• M OR P (1)<br>• NOT C (1)<br>• <expr1> AND <expr2> (1)<br><br>Example statements:<br><br>(M OR P) AND (NOT C)<br>(M OR P) AND NOT C<br>(M AND NOT C) OR (P AND NOT C)<br>M AND NOT C OR P AND NOT C<br><br>For information only:<br><br>
<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not<br><br>Order of precedence is NOT, AND, OR<br><br>Check brackets carefully, if provided</td><td>3</td></tr>
</table>

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(e) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. The operating system uses a scheduler to control processes (1)<br>2. The operating system holds processes in a queue (1)<br>3. Some processes may be given higher priorities than others (1)<br>4. Each process gains accesses to the CPU for a short time / time slice to execute (1)<br>5. Processes are swapped to/from (the queue / CPU) | Do not award responses about memory management, as the question is about sharing the CPU only | 2 |

## Type MS · Embedded systems: small computers inside devices

### Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Embedded systems: small computers inside devices

### Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a)(i) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• Heart rate (1) to measure blood flow/pulse rate (1)<br>• Barometric pressure (1) to determine altitude (1)<br>• Microphone (1) for voice activation control (1)<br>• Temperature/Thermometer (1) to measure ambient/body temperature (1)<br>• GPS (1) to calculate longitude/latitude (1)<br>• Light (1) to adjust screen brightness/charge watch battery (1)<br>• Magnetometer (1) to act as a compass (1)<br>• ECG (1) measure heart rhythm (1)<br>• Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1) | Allow any reasonable sensor that might be embedded within a smart watch.<br>Purpose must be linked to the sensor<br>Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark)<br>Allow purpose mentioned in second column if nothing in first column | **2** |
| Question | Answer | Additional Guidance | Mark |
| 5(a)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• It has a low power requirement (1)<br>• It can use a low processor speed (1)<br>• It has built in/integrated storage (1)<br>• It relies on flash memory (1)<br>• It can be mass produced at a low cost per device (1)<br>• It is small (1)<br>• It can work in a large range of operating temperature ranges (1)<br>• It is robust (1)<br>• It has its operating software set by the manufacturer (1)<br>• It is designed to do one particular task (1) | | **2** |

## Type MS · Embedded systems: small computers inside devices

### Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part a.i in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | **The only correct answer is B**<br><br>*A is not correct because low-level languages are not converted into high-level languages*<br>*C is not correct because assemblers translate assembly code to produce machine code but do not execute it*<br>*D is not correct because compilers and interpreters translate high-level languages* | | 1 |
| Question | Answer | Additional Guidance | Mark |
| 5(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1)<br>• It reduces manufacturing costs (1) because of a lower processor specification/expense (1)<br>• Optimise their code for hardware (1) directly access/control registers (1) | Do not accept references to translation time | 2 |
| Question | Answer | Additional Guidance | Mark |
| 5(b)(iii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• Abide by the law / avoid plagiarism / copyright infringements (1)<br>• Adhere to a professional code of conduct (1)<br>• Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1)<br>• Test code thoroughly/do not release code with known errors/vulnerabilities (1)<br>• Respect privacy and security of data (1)<br>• Preserve the environment (1)<br>• Make software accessible/unbiased (1) | | 2 |

## Type MS · Embedded systems: small computers inside devices

### Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part a.i in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(iv) | **The only correct answer is A**<br><br>*B is not correct because 500 KiB is 512,000 bytes*<br>*C is not correct because 0.5 MiB is 524,288 bytes*<br>*D is not correct because 5MB is 5,000,000 bytes* | | **1** |
| **Question** | **Answer** | **Additional Guidance** | **Mark** |
| 5(c) | Indicative content:<br><br>• Can use sensors to take/track readings such as heart rate<br>• Can track location via GPS and record routes taken/distance travelled<br>• Can calculate exercise/movement undertaken such as steps/calories burned<br>• Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data<br>• Can trigger alerts when users are inactive<br>• Can monitor blood oxygen/blood sugar/sleep patterns<br>• Can applaud/give kudos when an achievement/event takes place<br>• Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop<br>• Workouts/alerts can be synced to the watch<br>• Linked apps can perform more in-depth analysis<br>• No guarantee that users will buy into benefits and lead a healthier lifestyle<br>• Monitoring vulnerable people allowing independent living | Allow content referring to other wearable technology which is not a smartwatch | **6** |
| | | **Total for question 5** | **16** |

## Type MS · Embedded systems: small computers inside devices

### Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part a.i in this mark scheme extract

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1-2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

## Type MS · Embedded systems: small computers inside devices

### Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part a.i in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|